

# Nikhil Sinha

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## EDUCATION

### University of California, Los Angeles

*Bachelor of Science in Computer Science and Engineering*

Los Angeles, CA

Sept. 2026 – Jun. 2030

### American High School

*GPA: 3.98UW/4.57W*

Fremont, CA

Aug. 2022 – May 2026

## EXPERIENCE

### Founder and Team Captain

*Space Pickles Rocketry*

May 2024 – Apr. 2026

Fremont, CA

- Directed 28 members to launch 50+ rockets while competing in the American Rocketry Challenge (ARC).
- Developed full-stack internal inventory platform with React, PostgreSQL, and AWS for component tracking.
- Built sensor fusion and telemetry pipelines for state estimation with Kalman filtering; 11% apogee error reduction.
- Created trajectory simulation with MATLAB and nonlinear motion control algorithms in Python using CasADi.

### Software Vice Captain and Strategy Lead

*FIRST Robotics Competition (FRC) 6036 Peninsula Robotics*

Jun. 2024 – May 2026

Palo Alto, CA

- Reached 49/3700+ teams globally. Led archetype decisions, game analysis, and control system engineering.
- Developed inverse kinematics and nonlinear trajectory optimization solutions for 24% improved placement speed.
- Designed internal operations platform for team management with Node.js, Electron, and MongoDB.
- Contributed to camera and vision processing stack for AprilTag and object detection with OpenCV and YOLOv8.

### Founder and Developer

*iDENTify*

Oct. 2023 – Present

Fremont, CA

- Designed phone attachment in Onshape and AI-powered iOS app in Swift for remote oral healthcare monitoring.
- Trained PyTorch model detecting caries and gingivitis, achieving 89.7% accuracy annotating 4000+ images.
- Currently integrating with VELMEDI, raised 2K+ for POC, and working towards FDA approval.

### Software Engineering Intern

*TeachShare*

Jun. 2024 – Nov. 2024

Berkeley, CA

- Built chrome extension using Google App Script to generate worksheets and curriculums from user prompts.
- Worked on backend development employing Lexile-based text complexity analysis models and FastAPI.
- Engineered marketing automated workflows with REST API integrations, scaling platform to 20K+ users globally.

### Research Lead

*Aspiring Scholars Directed Research Program*

Aug. 2022 – Nov. 2025

Fremont, CA

- Presented transformer Alzheimer's model with longitudinal MRI tokens in PyTorch; 88% accuracy at SCCUR.
- Designed Dockerized AWS EC2 pipeline parallelizing preprocessing with FreeSurfer over 1,000+ MRI scans.
- Built autonomous drone system for rescue route optimization using SLAM, LiDAR, A\*, and Dijkstra's algorithm.

## PROJECTS

### Bundles | JavaScript, Supabase, Swift

Jun. 2026 – Present

- Developed iOS widget app in Expo and Supabase with real-time sync and APN push delivery for distant partners.
- Designed backend schema with Supabase with RLS and Realtime subscriptions for pairing and media delivery.

### 6-DOF Arm | Python, OpenCV, NumPy, pytest, MediaPipe

Sept. 2026 – Present

- Built control stack for COTS 5-DOF arm in Python with URDF-based IK to play handclap; sim-tested in CI.
- Brought up Opi5s for MediaPipe detection; reduced latency by 200ms with threaded loops and UDP across boards.

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript, HTML/CSS, TypeScript, Swift

**Frameworks/Technologies:** React, Node.js, Flask, Supabase, ROS, PyTorch, TensorFlow, YOLO, FastAPI

**Developer Tools:** Git, Docker, AWS, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ

**Libraries:** pandas, NumPy, SciPy, OpenCV, MATLAB, CasADi